sample_2-Evaluation report

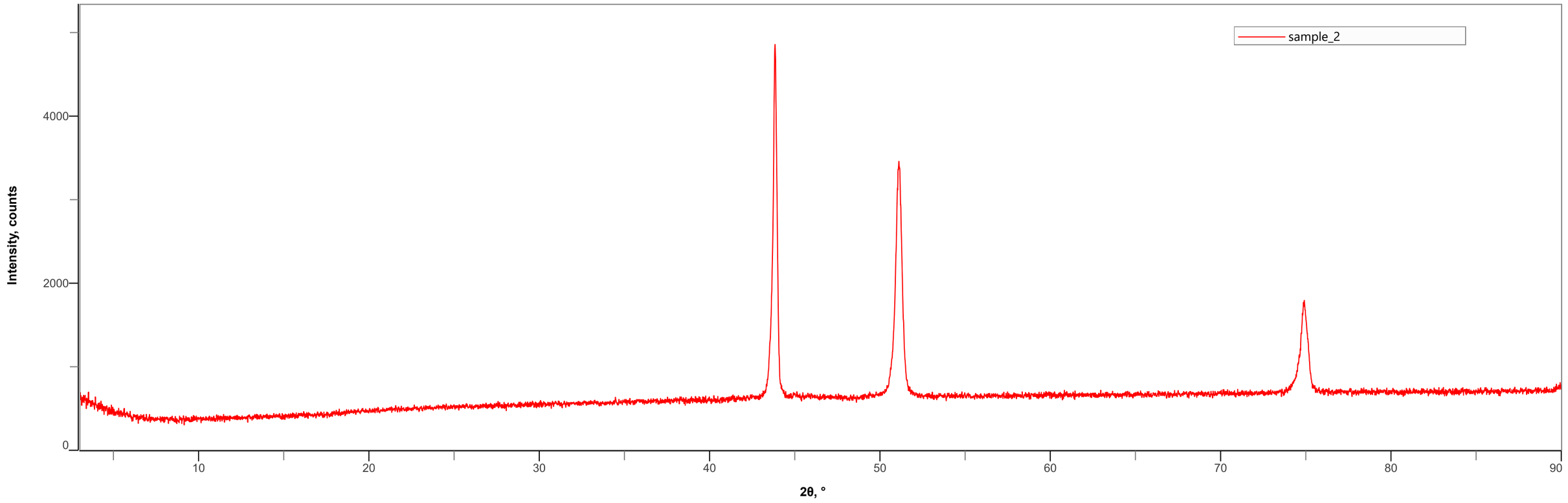
General information

Analysis date	2025-11-27 13:23:49	Measurement start time	2025-11-23 12:47:08
Analyst	Administrator	Operator	Administrator
Sample name	sample_2	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\sample_2.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	Standard sample stage	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

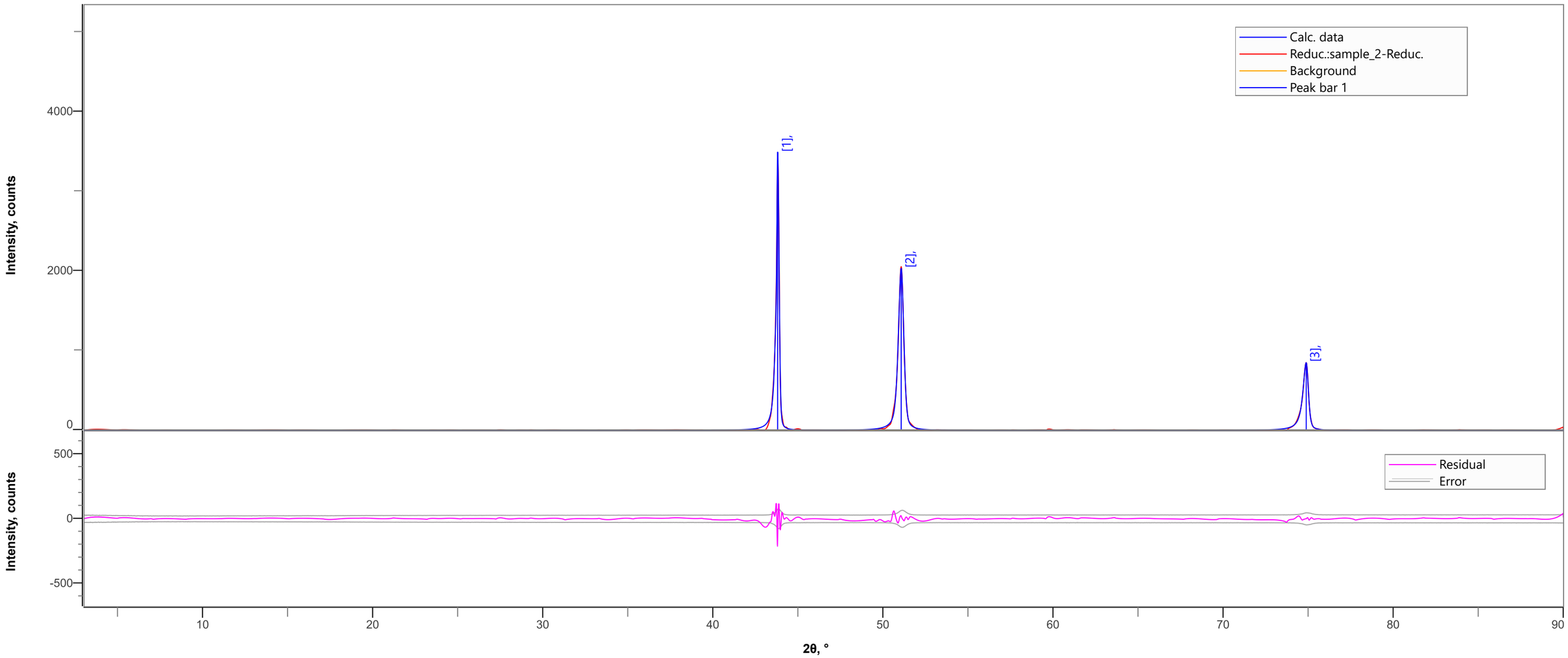
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	25.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

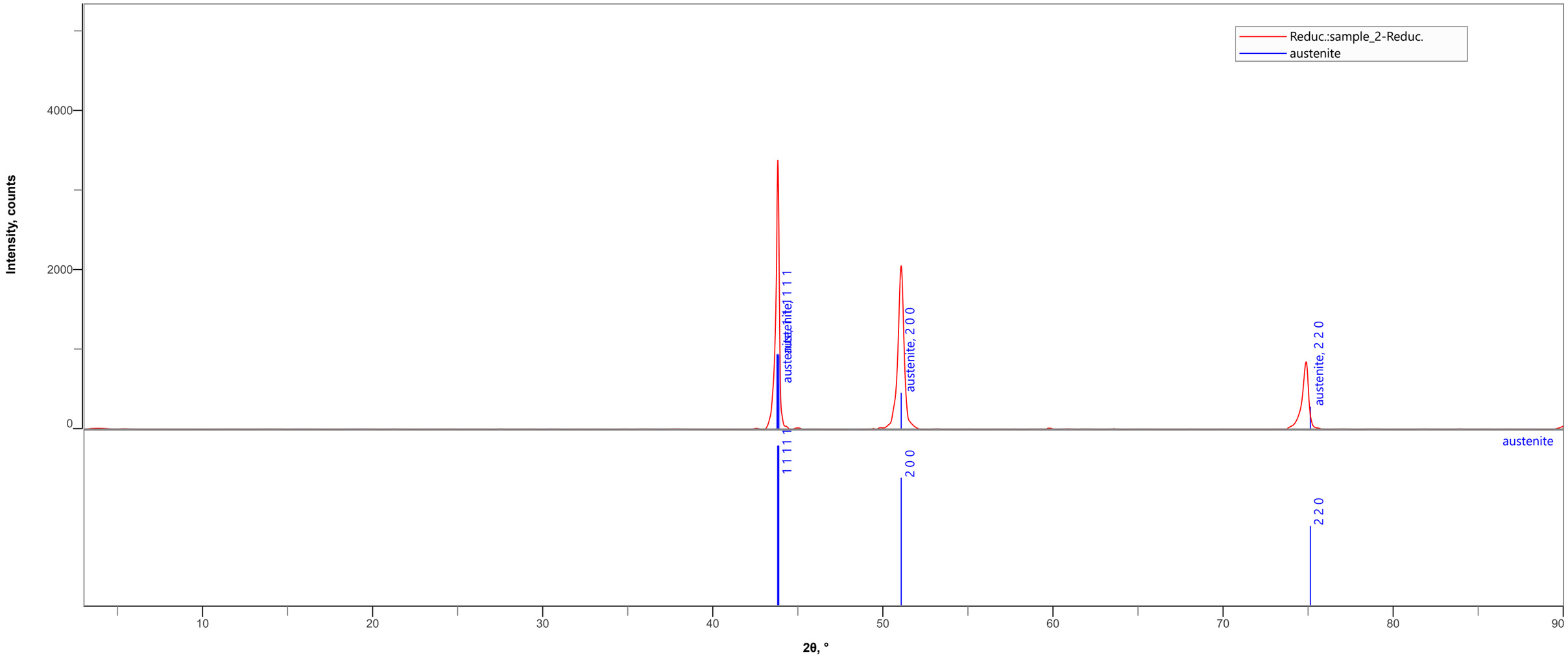
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
austenite	Cr - Ni - Fe - C	1.954	S/M:PDF-5+ 2026	225 : Fm-3m	00-031-0619

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, counts	FWHM, °	Int. I., counts°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay(ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	43.78866(15)	2.065714(7)	3488(77)	0.1766(3)	937.0(11)	0.269(6)	0.787(2)	1.409(3)	0.310(5)	506.2(8)	austenite: 1 1 1	Cr - Ni - Fe - C	00-031-0619	95.05	Split Pseudo-Voigt	-	-	-	-	-	
2	51.0494(3)	1.787643(10)	2031(63)	0.3714(6)	985.8(12)	0.485(16)	1.113(3)	0.663(4)	0.470(5)	247.5(4)	austenite: 2 0 0	Cr - Ni - Fe - C	00-031-0619	100.00	Split Pseudo-Voigt	-	-	-	-	-	
3	74.8465(6)	1.267564(8)	849(46)	0.3753(12)	451.7(10)	0.53(3)	1.496(9)	0.911(6)	0.533(11)	278.4(9)	austenite: 2 2 0	Cr - Ni - Fe - C	00-031-0619	45.82	Split Pseudo-Voigt	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
austenite	3.57529	3.57529	3.57529	90.000	90.000	90.000

d-I List

austenite

No.	2θ, °	d, Å	h k l	Norm. I.
1	43.82260	2.06419	1 1 1	100.00
2	51.04945	1.78764	2 0 0	80.00
3	75.09015	1.26405	2 2 0	50.00

Simple Inverse Pole Figure

austenite

